Mathematical Foundations of Political Methodology

Robert Gulotty

Associate Professor
Department of Political Science
University of Chicago

August 28th, 2017

- Vectors
- Matrix
- Matrix Algebra
- Matrix Inversion

- Vectors
- Matrix
- Matrix Algebra
- Matrix Inversion

Points and Vectors

Definition

A point $\boldsymbol{x} \in R^n$ *is an ordered n-tuple,* $(x_1, x_2, \ldots, x_n)$. *The vector* $\boldsymbol{x} \in R^n$ *is the arrow pointing from the origin* $(0, 0, \ldots, 0)$ *to* $\boldsymbol{x}$.

Some Examples:

- (Latitude, Longitude, Elevation)
- (1, 2, 3)
- (0, 1, 0)

Points and Vectors

Definition

A point $\boldsymbol{x} \in R^n$ *is an ordered n-tuple,* $(x_1, x_2, \ldots, x_n)$*. The vector* $\boldsymbol{x} \in R^n$ *is the arrow pointing from the origin* $(0, 0, \ldots, 0)$ *to* $\boldsymbol{x}$*.*

Some Examples:

- (Latitude, Longitude, Elevation)
- (1, 2, 3)
- (0, 1, 0)

Points and Vectors

Definition

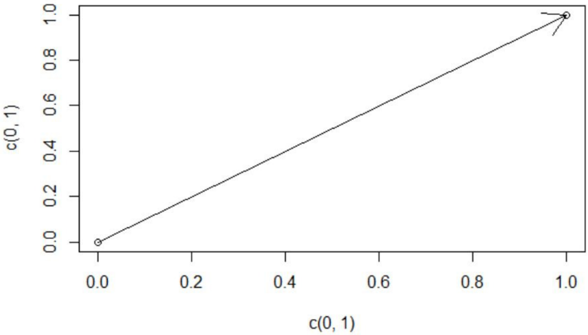
A point $\mathbf{x} \in R^n$ *is an ordered n-tuple,* $(x_1, x_2, \ldots, x_n)$*. The vector* $\mathbf{x} \in R^n$ *is the arrow pointing from the origin* $(0, 0, \ldots, 0)$ *to* $\mathbf{x}$*.*

Some Examples:

- (Latitude, Longitude, Elevation)
- (1, 2, 3)
- (0, 1, 0)

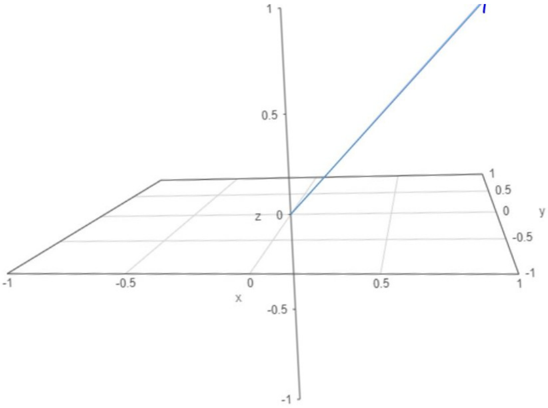
Points and Vectors

Now, let's see a vector in a 2 dimensional space.



Points and Vectors

Now, let's see a vector in a 3 dimensional space.



Arithmetic with Vectors

Definition

Suppose $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *are vectors* $\boldsymbol{u}, \boldsymbol{v} \in R^n$,

$$\begin{aligned} \boldsymbol{u} &= (u_1, u_2, u_3, \ldots, u_n) \\ \boldsymbol{v} &= (v_1, v_2, v_3, \ldots, v_n) \end{aligned}$$

We will say $\boldsymbol{u} = \boldsymbol{v}$ *if* $u_1 = v_1, u_2 = v_2, \ldots, u_n = v_n$
Define the sum of $\boldsymbol{u} + \boldsymbol{v}$ *as*

$$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3, \ldots, u_n + v_n)$$

Suppose $k \in R$. *We will call* k *a scalar.*
Define $k\boldsymbol{u}$ *as the scalar product*

$$k\boldsymbol{u} = (ku_1, ku_2, \ldots, ku_n)$$

Arithmetic with Vectors

Definition

Suppose $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *are vectors* $\boldsymbol{u}, \boldsymbol{v} \in R^n$,

$$\begin{aligned} \boldsymbol{u} &= (u_1, u_2, u_3, \ldots, u_n) \\ \boldsymbol{v} &= (v_1, v_2, v_3, \ldots, v_n) \end{aligned}$$

We will say $\boldsymbol{u} = \boldsymbol{v}$ *if* $u_1 = v_1, u_2 = v_2, \ldots, u_n = v_n$
Define the sum of $\boldsymbol{u} + \boldsymbol{v}$ *as*

$$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3, \ldots, u_n + v_n)$$

Suppose $k \in R$. *We will call* k *a scalar.*
Define $k\boldsymbol{u}$ *as the scalar product*

$$k\boldsymbol{u} = (ku_1, ku_2, \ldots, ku_n)$$

Arithmetic with Vectors

Definition

Suppose $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *are vectors* $\boldsymbol{u}, \boldsymbol{v} \in R^n$*,*

$$\begin{aligned} \boldsymbol{u} &= (u_1, u_2, u_3, \ldots, u_n) \\ \boldsymbol{v} &= (v_1, v_2, v_3, \ldots, v_n) \end{aligned}$$

We will say $\boldsymbol{u} = \boldsymbol{v}$ *if* $u_1 = v_1, u_2 = v_2, \ldots, u_n = v_n$
Define the sum of $\boldsymbol{u} + \boldsymbol{v}$ *as*

$$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3, \ldots, u_n + v_n)$$

Suppose $k \in R$*. We will call* k *a scalar.*
Define $k\boldsymbol{u}$ *as the scalar product*

$$k\boldsymbol{u} = (ku_1, ku_2, \ldots, ku_n)$$

Examples

Suppose:

$$\begin{aligned} \boldsymbol{u} &= (1,2,3,4,5) \\ \boldsymbol{v} &= (1,1,1,1,1) \\ k &= 2 \end{aligned}$$

Then,

$$\begin{aligned} \boldsymbol{u}+\boldsymbol{v} &= (1+1,2+1,3+1,4+1,5+1) = (2,3,4,5,6) \\ k\boldsymbol{u} &= (2\times 1,2\times 2,2\times 3,2\times 4,2\times 5) = (2,4,6,8,10) \\ k\boldsymbol{v} &= (2\times 1,2\times 1,2\times 1,2\times 1,2\times 1) = (2,2,2,2,2) \end{aligned}$$

Examples

Suppose:

$$\begin{aligned} \boldsymbol{u} &= (1,2,3,4,5) \\ \boldsymbol{v} &= (1,1,1,1,1) \\ k &= 2 \end{aligned}$$

Then,

$$\begin{aligned} \boldsymbol{u}+\boldsymbol{v} &= (1+1,2+1,3+1,4+1,5+1) = (2,3,4,5,6) \\ k\boldsymbol{u} &= (2\times 1, 2\times 2, 2\times 3, 2\times 4, 2\times 5) = (2,4,6,8,10) \\ k\boldsymbol{v} &= (2\times 1, 2\times 1, 2\times 1, 2\times 1, 2\times 1) = (2,2,2,2,2) \end{aligned}$$

Examples

Suppose:

$$\begin{aligned} \boldsymbol{u} &= (1, 2, 3, 4, 5) \\ \boldsymbol{v} &= (1, 1, 1, 1, 1) \\ k &= 2 \end{aligned}$$

Then,

$$\begin{aligned} \boldsymbol{u} + \boldsymbol{v} &= (1+1, 2+1, 3+1, 4+1, 5+1) = (2, 3, 4, 5, 6) \\ k\boldsymbol{u} &= (2 \times 1, 2 \times 2, 2 \times 3, 2 \times 4, 2 \times 5) = (2, 4, 6, 8, 10) \\ k\boldsymbol{v} &= (2 \times 1, 2 \times 1, 2 \times 1, 2 \times 1, 2 \times 1) = (2, 2, 2, 2, 2) \end{aligned}$$

Examples

Suppose:

$$\begin{aligned} \boldsymbol{u} &= (1,2,3,4,5) \\ \boldsymbol{v} &= (1,1,1,1,1) \\ k &= 2 \end{aligned}$$

Then,

$$\begin{aligned} \boldsymbol{u}+\boldsymbol{v} &= (1+1,2+1,3+1,4+1,5+1) = (2,3,4,5,6) \\ k\boldsymbol{u} &= (2\times 1,2\times 2,2\times 3,2\times 4,2\times 5) = (2,4,6,8,10) \\ k\boldsymbol{v} &= (2\times 1,2\times 1,2\times 1,2\times 1,2\times 1) = (2,2,2,2,2) \end{aligned}$$

Euclidean Norm of a Vector

Now, a question will be: how do we measure the length of a vector.

One way is to use Euclidean norm to measure the length of a vector

Definition

On an n-dimensional Euclidean space R^n, the Euclidean norm of vector, $x = (x_1, x_2, ..., x_n)$, is defined by the formula

$$||x||_2 := \sqrt{x_1^2 + \cdots + x_n^2}.$$

An example. What's the Euclidean norm of $x = (1, 3, 5)$?

Its Euclidean norm equals to:

$$\sqrt{1^2 + 3^2 + 5^2} = \sqrt{35}$$

.

Euclidean Norm of a Vector

Now, a question will be: how do we measure the length of a vector.

One way is to use Euclidean norm to measure the length of a vector

Definition

On an n-dimensional Euclidean space R^n, the Euclidean norm of vector, $x = (x_1, x_2, ..., x_n)$, is defined by the formula

$$||x||_2 := \sqrt{x_1^2 + \cdots + x_n^2}.$$

An example. What's the Euclidean norm of $x = (1, 3, 5)$?

Its Euclidean norm equals to:

$$\sqrt{1^2 + 3^2 + 5^2} = \sqrt{35}$$

.

Euclidean Norm of a Vector

Now, a question will be: how do we measure the length of a vector.

One way is to use Euclidean norm to measure the length of a vector

Definition

On an n-dimensional Euclidean space R^n, the Euclidean norm of vector, $x = (x_1, x_2, ..., x_n)$, is defined by the formula

$$||x||_2 := \sqrt{x_1^2 + \cdots + x_n^2}.$$

An example. What's the Euclidean norm of $x = (1, 3, 5)$?

Its Euclidean norm equals to:

$$\sqrt{1^2 + 3^2 + 5^2} = \sqrt{35}$$

.

Euclidean Norm of a Vector

Now, a question will be: how do we measure the length of a vector.

One way is to use Euclidean norm to measure the length of a vector

Definition

On an n-dimensional Euclidean space R^n, the Euclidean norm of vector, $x = (x_1, x_2, ..., x_n)$, is defined by the formula

$$||x||_2 := \sqrt{x_1^2 + \cdots + x_n^2}.$$

An example. What's the Euclidean norm of $x = (1, 3, 5)$?

Its Euclidean norm equals to:

$$\sqrt{1^2 + 3^2 + 5^2} = \sqrt{35}$$

.

- Vectors
- Matrix
- Matrix Algebra
- Matrix Inversion

Matrices

Now, let's see some examples for using matrices.

The first example is the following system of equations:

$$\begin{aligned} 2x + 2y + 3z &= 1 \\ 3x + 5y + 7z &= 2 \\ 5x + 8y + 3z &= 3 \end{aligned}$$

The coefficient of unknowns can be written compactly as matrices:
$$\begin{bmatrix} 2 & 2 & 3 \\ 3 & 5 & 7 \\ 5 & 8 & 3 \end{bmatrix}$$

Matrices

Now, let's see some examples for using matrices.

The first example is the following system of equations:

$$
\begin{aligned}
2x + 2y + 3z &= 1 \\
3x + 5y + 7z &= 2 \\
5x + 8y + 3z &= 3
\end{aligned}
$$

The coefficient of unknowns can be written compactly as matrices:

$$
\begin{bmatrix}
2 & 2 & 3 \\
3 & 5 & 7 \\
5 & 8 & 3
\end{bmatrix}
$$

Matrices

Now, let's see some examples for using matrices.

The first example is the following system of equations:

$$\begin{aligned} 2x + 2y + 3z &= 1 \\ 3x + 5y + 7z &= 2 \\ 5x + 8y + 3z &= 3 \end{aligned}$$

The coefficient of unknowns can be written compactly as matrices:

$$\begin{bmatrix} 2 & 2 & 3 \\ 3 & 5 & 7 \\ 5 & 8 & 3 \end{bmatrix}$$

Matrices

Now, let's see some examples for using matrices.

The first example is the following system of equations:

$$\begin{aligned} 2x + 2y + 3z &= 1 \\ 3x + 5y + 7z &= 2 \\ 5x + 8y + 3z &= 3 \end{aligned}$$

The coefficient of unknowns can be written compactly as matrices:

$$\begin{bmatrix} 2 & 2 & 3 \\ 3 & 5 & 7 \\ 5 & 8 & 3 \end{bmatrix}$$

Matrices

A second example of matrix is the data set.

	id	name	year	urbanpop	totalpop	urbrate	fpresence	westelbe	protestant	latitude	longitude
1	1	Rhineland	1700	83000	.	.	19	1	.25	50.35	7.6
2	1	Rhineland	1750	129000	1438817	8.9657	19	1	.25	50.35	7.6
3	1	Rhineland	1800	274000	1860719	14.72549	19	1	.25	50.35	7.6
4	1	Rhineland	1850	572278	2846708	20.10317	19	1	.25	50.35	7.6
5	1	Rhineland	1875	1629051	3804381	42.8204	19	1	.25	50.35	7.6
6	1	Rhineland	1880	1850627	4074000	45.42531	19	1	.25	50.35	7.6
7	1	Rhineland	1885	2103753	4344527	48.42306	19	1	.25	50.35	7.6
8	1	Rhineland	1895	2836140	5106002	55.54522	19	1	.25	50.35	7.6
9	1	Rhineland	1900	3506587	5759798	60.88038	19	1	.25	50.35	7.6
10	1	Rhineland	1905	4145633	6436337	64.40982	19	1	.25	50.35	7.6
11	1	Rhineland	1910	4806725	7121140	67.49937	19	1	.25	50.35	7.6
12	2	Mark	1700	5000	117963	4.238618	6	1	.6	51.68333	7.816667
13	2	Mark	1750	5000	149869	3.336254	6	1	.6	51.68333	7.816667
14	2	Mark	1800	36000	193096	18.64353	6	1	.6	51.68333	7.816667
15	2	Mark	1850	55628	362392	15.35011	6	1	.6	51.68333	7.816667
16	2	Mark	1875	305301	721466	42.31675	6	1	.6	51.68333	7.816667
17	2	Mark	1880	.	.	.	6	1	.6	51.68333	7.816667
18	2	Mark	1885	.	.	.	6	1	.6	51.68333	7.816667
19	2	Mark	1895	.	.	.	6	1	.6	51.68333	7.816667
20	2	Mark	1900	1134655	1514546	74.91717	6	1	.6	51.68333	7.816667
21	2	Mark	1905	.	.	.	6	1	.6	51.68333	7.816667
22	2	Mark	1910	.	.	.	6	1	.6	51.68333	7.816667
23	3	Westphalia without Mark	1700	7000	.	.	6	1	.4	51.96667	7.633333
24	3	Westphalia without Mark	1750	14000	528851	2.647246	6	1	.4	51.96667	7.633333
25	3	Westphalia without Mark	1800	55000	763110	7.20735	6	1	.4	51.96667	7.633333
26	3	Westphalia without Mark	1850	61000	1120946	5.441833	6	1	.4	51.96667	7.633333
27	3	Westphalia without Mark	1875	190503	1184231	16.08664	6	1	.4	51.96667	7.633333
28	3	Westphalia without Mark	1880	.	.	.	6	1	.4	51.96667	7.633333

Matrices

Definition

A Matrix is a rectangular array of numbers

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

If $\boldsymbol{A}$ *has m rows n columns (Roman Catholic; Rum & Cake) we will say that* $\boldsymbol{A}$ *is an* $m \times n$ *matrix.*
Suppose $\boldsymbol{X}$ *and* $\boldsymbol{Y}$ *are* $m \times n$ *matrices. Then* $\boldsymbol{X} = \boldsymbol{Y}$ *if* $x_{ij} = y_{ij}$ *for all i and j*

Simple Examples

$$\boldsymbol{I} \quad = \quad \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

If $\boldsymbol{I}$ is an $n \times n$ matrix we will call an identity matrix.

Simple Examples

$$\boldsymbol{X} \;=\; \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \end{pmatrix}$$

$\boldsymbol{X}$ is an 2×3 matrix

- Vectors
- Matrix
- Matrix Algebra
- Matrix Inversion

Matrix Algebra

Definition

Suppose $\boldsymbol{X}$ *and* $\boldsymbol{Y}$ *are* $m \times n$ *matrices. Then define*

$$
\begin{aligned}
\boldsymbol{X} + \boldsymbol{Y} &= \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix} + \begin{pmatrix} y_{11} & y_{12} & \dots & y_{1n} \\ y_{21} & y_{22} & \dots & y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{m1} & y_{m2} & \dots & y_{mn} \end{pmatrix} \\
&= \begin{pmatrix} x_{11} + y_{11} & x_{12} + y_{12} & \dots & x_{1n} + y_{1n} \\ x_{21} + y_{21} & x_{22} + y_{22} & \dots & x_{2n} + y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} + y_{m1} & x_{m2} + y_{m2} & \dots & x_{mn} + y_{mn} \end{pmatrix}
\end{aligned}
$$

Matrix Algebra

Definition

Suppose $\boldsymbol{X}$ *is an* $m \times n$ *matrix and* $k \in R$. *Then,*

$$k\boldsymbol{X} = \begin{pmatrix} kx_{11} & kx_{12} & \dots & kx_{1n} \\ kx_{21} & kx_{22} & \dots & kx_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ kx_{m1} & kx_{m2} & \dots & kx_{mn} \end{pmatrix}$$

Matrix Transpose

We will use matrix transpose to flip the dimensionality of a matrix

$$\boldsymbol{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix}$$

$$\boldsymbol{X}' = \begin{pmatrix} x_{11} & x_{21} & \cdots & x_{m1} \\ x_{12} & x_{22} & \cdots & x_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \cdots & x_{mn} \end{pmatrix}$$

If $\boldsymbol{X}$ is an $m \times n$ then $\boldsymbol{X}'$ is $n \times m$.
If $\boldsymbol{X} = \boldsymbol{X}'$ then we say $\boldsymbol{X}$ is symmetric.

Matrix Transpose

We will use matrix transpose to flip the dimensionality of a matrix

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$$

$$\mathbf{X}' = \begin{pmatrix} x_{11} & x_{21} & \dots & x_{m1} \\ x_{12} & x_{22} & \dots & x_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \dots & x_{mn} \end{pmatrix}$$

If $\mathbf{X}$ is an $m \times n$ then $\mathbf{X}'$ is $n \times m$.
If $\mathbf{X} = \mathbf{X}'$ then we say $\mathbf{X}$ is symmetric.

Matrix Transpose

We will use matrix transpose to flip the dimensionality of a matrix

$$\boldsymbol{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix}$$

$$\boldsymbol{X}^{'} = \begin{pmatrix} x_{11} & x_{21} & \cdots & x_{m1} \\ x_{12} & x_{22} & \cdots & x_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \cdots & x_{mn} \end{pmatrix}$$

If $\boldsymbol{X}$ is an $m \times n$ then $\boldsymbol{X}^{'}$ is $n \times m$.
If $\boldsymbol{X} = \boldsymbol{X}^{'}$ then we say $\boldsymbol{X}$ is symmetric.

Matrix Transpose

We will use matrix transpose to flip the dimensionality of a matrix

$$
\begin{aligned}
\boldsymbol{X} &= \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix} \\
\boldsymbol{X}^{'} &= \begin{pmatrix} x_{11} & x_{21} & \dots & x_{m1} \\ x_{12} & x_{22} & \dots & x_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \dots & x_{mn} \end{pmatrix}
\end{aligned}
$$

If $\boldsymbol{X}$ is an $m \times n$ then $\boldsymbol{X}^{'}$ is $n \times m$.
If $\boldsymbol{X} = \boldsymbol{X}^{'}$ then we say $\boldsymbol{X}$ is symmetric.

Matrix Transpose

We will use matrix transpose to flip the dimensionality of a matrix

$$\boldsymbol{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$$

$$\boldsymbol{X}^{'} = \begin{pmatrix} x_{11} & x_{21} & \dots & x_{m1} \\ x_{12} & x_{22} & \dots & x_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \dots & x_{mn} \end{pmatrix}$$

If $\boldsymbol{X}$ is an $m \times n$ then $\boldsymbol{X}^{'}$ is $n \times m$.
If $\boldsymbol{X} = \boldsymbol{X}^{'}$ then we say $\boldsymbol{X}$ is symmetric.

Matrix Transpose

We will use matrix transpose to flip the dimensionality of a matrix

$$\boldsymbol{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$$

$$\boldsymbol{X}^{'} = \begin{pmatrix} x_{11} & x_{21} & \dots & x_{m1} \\ x_{12} & x_{22} & \dots & x_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \dots & x_{mn} \end{pmatrix}$$

If $\boldsymbol{X}$ is an $m \times n$ then $\boldsymbol{X}^{'}$ is $n \times m$.
If $\boldsymbol{X} = \boldsymbol{X}^{'}$ then we say $\boldsymbol{X}$ is symmetric.

Matrix Transpose

Example 1: $\boldsymbol{X} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$ then $\boldsymbol{X}' = \begin{pmatrix} 4 & 1 \\ 1 & 2 \\ 2 & 3 \end{pmatrix}$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$
\begin{aligned}
\boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix}
\end{aligned}
$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$\begin{aligned}
\boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix}
\end{aligned}$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$\begin{aligned} \boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\ &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix} \end{aligned}$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$\begin{aligned} \boldsymbol{A} &= \boldsymbol{XY} \\ &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 1\times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1\times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix} \end{aligned}$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$\begin{aligned} \boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\ &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix} \end{aligned}$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$\begin{aligned}
\boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix}
\end{aligned}$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$\begin{aligned} \boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\ &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix} \end{aligned}$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$
\begin{aligned}
\boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix}
\end{aligned}
$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$
\begin{aligned}
\boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix}
\end{aligned}
$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$
\begin{aligned}
\boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix}
\end{aligned}
$$

Matrix Multiplication

How do we multiply matrices?

Because we want to use matrix multiplication to solve equations we won't use an intuitive definition

Suppose we have two matrices

$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

We will create a new matrix $\boldsymbol{A}$ by matrix multiplication:

$$
\begin{aligned}
\boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix}
\end{aligned}
$$

Matrix Multiplication

Definition

Suppose $\boldsymbol{X}$ *is an* $m \times n$ *matrix and* $\boldsymbol{Y}$ *is an* $n \times k$ *matrix. Then define the matrix* $\boldsymbol{A}$ *an* $m \times k$ *matrix that obtains from multiplying* $\boldsymbol{X}$ *and* $\boldsymbol{Y}$ *as,*

$$
\begin{aligned}
\boldsymbol{A} &= \boldsymbol{X}\boldsymbol{Y} \\
&= \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix} \begin{pmatrix} y_{11} & y_{12} & \dots & y_{1k} \\ y_{21} & y_{22} & \dots & y_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \dots & y_{nk} \end{pmatrix} \\
&= \begin{pmatrix} x_{11}y_{11} + x_{12}y_{21} + \ldots + x_{1n}y_{n1} & \dots & x_{11}y_{1k} + x_{12}y_{2k} + \ldots + x_{1n}y_{nk} \\ \vdots & \ddots & \vdots \\ x_{m1}y_{11} + x_{m2}y_{21} + \ldots + x_{mn}y_{n1} & \dots & x_{m1}y_{11} + x_{m2}y_{12} + \ldots + x_{mn}y_{nk} \end{pmatrix}
\end{aligned}
$$

Definition

Suppose $\boldsymbol{X}$ *is an* $m \times n$ *matrix and* $\boldsymbol{Y}$ *is an* $n \times k$ *matrix. Write the row vectors of* $\boldsymbol{X} = \begin{pmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \\ \vdots \\ \boldsymbol{x}_m \end{pmatrix}$ *and* $\boldsymbol{Y}$ *as column vector* $\boldsymbol{Y} = \begin{pmatrix} \boldsymbol{y}_1 & \boldsymbol{y}_2 & \dots & \boldsymbol{y}_k \end{pmatrix}$.

Then the $m \times k$ *matrix* $\boldsymbol{A} = \boldsymbol{X}\boldsymbol{Y}$ *can be written as*

$$\boldsymbol{A} = \begin{pmatrix} \boldsymbol{x}_1 \cdot \boldsymbol{y}_1 & \boldsymbol{x}_1 \cdot \boldsymbol{y}_2 & \dots & \boldsymbol{x}_1 \cdot \boldsymbol{y}_k \\ \boldsymbol{x}_2 \cdot \boldsymbol{y}_1 & \boldsymbol{x}_2 \cdot \boldsymbol{y}_2 & \dots & \boldsymbol{x}_2 \cdot \boldsymbol{y}_k \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{x}_m \cdot \boldsymbol{y}_1 & \boldsymbol{x}_m \cdot \boldsymbol{y}_2 & \dots & \boldsymbol{x}_m \cdot \boldsymbol{y}_k \end{pmatrix}$$

Matrix Multiplication

Let's work on an example together!

$\boldsymbol{X} = \begin{pmatrix} 1 & 4 & 5 \\ 10 & 2 & 3 \end{pmatrix}$

$\boldsymbol{Y} = \begin{pmatrix} 2 & 3 \\ 1 & 5 \\ 3 & 5 \end{pmatrix}$ What is $\boldsymbol{XY}$?

Not all matrices can be multiplied.

Matrix $\boldsymbol{AB}$ exists only if the number of columns in $\boldsymbol{A}$ = number of rows in $\boldsymbol{B}$. If $\boldsymbol{AB}$ exists we will say the matrices are conformable

Matrix Multiplication

Let's work on an example together!

$\boldsymbol{X} = \begin{pmatrix} 1 & 4 & 5 \\ 10 & 2 & 3 \end{pmatrix}$

$\boldsymbol{Y} = \begin{pmatrix} 2 & 3 \\ 1 & 5 \\ 3 & 5 \end{pmatrix}$ What is $\boldsymbol{XY}$?

Not all matrices can be multiplied.

Matrix $\boldsymbol{AB}$ exists only if the number of columns in $\boldsymbol{A}$ = number of rows in $\boldsymbol{B}$. If $\boldsymbol{AB}$ exists we will say the matrices are conformable

Matrix Multiplication with a Vector

Suppose $\boldsymbol{X} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 1 & 5 & 1 & 2 \\ 3 & 5 & 3 & 4 \end{pmatrix}$ a 3×4 matrix and that $\boldsymbol{v} = \begin{pmatrix} 3 \\ 3 \\ 4 \\ 10 \end{pmatrix}$ a 4×1 matrix (or a column vector) what is
$\boldsymbol{X}\boldsymbol{v}$?
What is $\boldsymbol{X}^{'}\boldsymbol{v}$?

Algebraic Properties

Suppose $\boldsymbol{X}$ is an $m \times n$ matrix and $\boldsymbol{Y}$ is an $n \times k$ matrix. Suppose that
$\boldsymbol{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$ as the identity matrix and that $k \in R$.

- $\boldsymbol{XY} \neq \boldsymbol{YX}$ in general
- $\boldsymbol{XI} = \boldsymbol{X}$
- $(\boldsymbol{X}')' = \boldsymbol{X}$
- $(\boldsymbol{XY})' = \boldsymbol{Y}'\boldsymbol{X}'$
- $(k\boldsymbol{X})' = k\boldsymbol{X}'$
- $(\boldsymbol{X} + \boldsymbol{Y})' = \boldsymbol{X}' + \boldsymbol{Y}'$

Algebraic Properties

Suppose $\boldsymbol{X}$ is an $m \times n$ matrix and $\boldsymbol{Y}$ is an $n \times k$ matrix. Suppose that

$$\boldsymbol{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

as the identity matrix and that $k \in R$.

- $\boldsymbol{XY} \neq \boldsymbol{YX}$ in general
- $\boldsymbol{XI} = \boldsymbol{X}$
- $(\boldsymbol{X}')' = \boldsymbol{X}$
- $(\boldsymbol{XY})' = \boldsymbol{Y}'\boldsymbol{X}'$
- $(k\boldsymbol{X})' = k\boldsymbol{X}'$
- $(\boldsymbol{X} + \boldsymbol{Y})' = \boldsymbol{X}' + \boldsymbol{Y}'$

Algebraic Properties

Suppose $\boldsymbol{X}$ is an $m \times n$ matrix and $\boldsymbol{Y}$ is an $n \times k$ matrix. Suppose that

$\boldsymbol{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$ as the identity matrix and that $k \in R$.

- $\boldsymbol{XY} \neq \boldsymbol{YX}$ in general
- $\boldsymbol{XI} = \boldsymbol{X}$
- $(\boldsymbol{X}')' = \boldsymbol{X}$
- $(\boldsymbol{XY})' = \boldsymbol{Y}'\boldsymbol{X}'$
- $(k\boldsymbol{X})' = k\boldsymbol{X}'$
- $(\boldsymbol{X} + \boldsymbol{Y})' = \boldsymbol{X}' + \boldsymbol{Y}'$

Algebraic Properties

Suppose $\boldsymbol{X}$ is an $m \times n$ matrix and $\boldsymbol{Y}$ is an $n \times k$ matrix. Suppose that $\boldsymbol{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$ as the identity matrix and that $k \in R$.

- $\boldsymbol{X}\boldsymbol{Y} \neq \boldsymbol{Y}\boldsymbol{X}$ in general
- $\boldsymbol{X}\boldsymbol{I} = \boldsymbol{X}$
- $(\boldsymbol{X}')' = \boldsymbol{X}$
- $(\boldsymbol{X}\boldsymbol{Y})' = \boldsymbol{Y}'\boldsymbol{X}'$
- $(k\boldsymbol{X})' = k\boldsymbol{X}'$
- $(\boldsymbol{X} + \boldsymbol{Y})' = \boldsymbol{X}' + \boldsymbol{Y}'$

Algebraic Properties

Suppose $\boldsymbol{X}$ is an $m \times n$ matrix and $\boldsymbol{Y}$ is an $n \times k$ matrix. Suppose that $\boldsymbol{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$ as the identity matrix and that $k \in R$.

- $\boldsymbol{XY} \neq \boldsymbol{YX}$ in general
- $\boldsymbol{XI} = \boldsymbol{X}$
- $(\boldsymbol{X}')' = \boldsymbol{X}$
- $(\boldsymbol{XY})' = \boldsymbol{Y}'\boldsymbol{X}'$
- $(k\boldsymbol{X})' = k\boldsymbol{X}'$
- $(\boldsymbol{X} + \boldsymbol{Y})' = \boldsymbol{X}' + \boldsymbol{Y}'$

Algebraic Properties

Suppose $\boldsymbol{X}$ is an $m \times n$ matrix and $\boldsymbol{Y}$ is an $n \times k$ matrix. Suppose that $\boldsymbol{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$ as the identity matrix and that $k \in R$.

- $\boldsymbol{XY} \neq \boldsymbol{YX}$ in general
- $\boldsymbol{XI} = \boldsymbol{X}$
- $(\boldsymbol{X}')' = \boldsymbol{X}$
- $(\boldsymbol{XY})' = \boldsymbol{Y}'\boldsymbol{X}'$
- $(k\boldsymbol{X})' = k\boldsymbol{X}'$
- $(\boldsymbol{X} + \boldsymbol{Y})' = \boldsymbol{X}' + \boldsymbol{Y}'$

Algebraic Properties

Suppose $\boldsymbol{X}$ is an $m \times n$ matrix and $\boldsymbol{Y}$ is an $n \times k$ matrix. Suppose that

$$\boldsymbol{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$ as the identity matrix and that $k \in R$.

- $\boldsymbol{XY} \neq \boldsymbol{YX}$ in general
- $\boldsymbol{XI} = \boldsymbol{X}$
- $(\boldsymbol{X}^{'})^{'} = \boldsymbol{X}$
- $(\boldsymbol{XY})^{'} = \boldsymbol{Y}^{'}\boldsymbol{X}^{'}$
- $(k\boldsymbol{X})^{'} = k\boldsymbol{X}^{'}$
- $(\boldsymbol{X} + \boldsymbol{Y})^{'} = \boldsymbol{X}^{'} + \boldsymbol{Y}^{'}$

- Vectors
- Matrix
- Matrix Algebra
- Matrix Inversion

Matrix Inversion: Motivations

Big topic: suppose $\boldsymbol{X}$ is an $n \times n$ matrix. We want to find the matrix X^{-1} such that

$$\boldsymbol{X}^{-1}\boldsymbol{X} \;=\; \boldsymbol{X}\boldsymbol{X}^{-1} = \boldsymbol{I}$$

where $\boldsymbol{I}$ is the $n \times n$ identity matrix.

Why do we care about matrix inversion?

It will help us to determine whether a system of equations is solvable.

Matrix Inversion: Motivations

Big topic: suppose $\boldsymbol{X}$ is an $n \times n$ matrix. We want to find the matrix X^{-1} such that

$$\boldsymbol{X}^{-1}\boldsymbol{X} \;=\; \boldsymbol{X}\boldsymbol{X}^{-1} = \boldsymbol{I}$$

where $\boldsymbol{I}$ is the $n \times n$ identity matrix.

Why do we care about matrix inversion?

It will help us to determine whether a system of equations is solvable.

Matrix Inversion: Motivations

Big topic: suppose $\boldsymbol{X}$ is an $n \times n$ matrix. We want to find the matrix X^{-1} such that

$$\boldsymbol{X}^{-1}\boldsymbol{X} \;=\; \boldsymbol{X}\boldsymbol{X}^{-1} = \boldsymbol{I}$$

where $\boldsymbol{I}$ is the $n \times n$ identity matrix.

Why do we care about matrix inversion?

It will help us to determine whether a system of equations is solvable.

Matrix Inversion: Motivations

Consider the following equations:

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ 0x_1 + 5x_2 + 0x_3 &= 5 \\ 0x_1 + 0x_2 + 3x_3 &= 6 \end{aligned}$$

$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

$\boldsymbol{x} = (x_1, x_2, x_3)$

$\boldsymbol{b} = (0, 5, 6)$

The system of equations are now,

$$\boldsymbol{Ax} = \boldsymbol{b}$$

$\boldsymbol{A}^{-1}$ exists if and only if $\boldsymbol{Ax} = \boldsymbol{b}$ has only one solution.

Matrix Inversion: Motivations

Consider the following equations:

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ 0x_1 + 5x_2 + 0x_3 &= 5 \\ 0x_1 + 0x_2 + 3x_3 &= 6 \end{aligned}$$

$$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$\boldsymbol{x} = (x_1, x_2, x_3)$

$\boldsymbol{b} = (0, 5, 6)$

The system of equations are now,

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$$

$\boldsymbol{A}^{-1}$ exists if and only if $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ has only one solution.

Matrix Inversion: Motivations

Consider the following equations:

$$
\begin{aligned}
x_1 + x_2 + x_3 &= 0 \\
0x_1 + 5x_2 + 0x_3 &= 5 \\
0x_1 + 0x_2 + 3x_3 &= 6
\end{aligned}
$$

$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

$\boldsymbol{x} = (x_1, x_2, x_3)$

$\boldsymbol{b} = (0, 5, 6)$

The system of equations are now,

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$$

$\boldsymbol{A}^{-1}$ exists if and only if $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ has only one solution.

Matrix Inversion: Motivations

Consider the following equations:

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ 0x_1 + 5x_2 + 0x_3 &= 5 \\ 0x_1 + 0x_2 + 3x_3 &= 6 \end{aligned}$$

$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$
$\boldsymbol{x} = (x_1, x_2, x_3)$
$\boldsymbol{b} = (0, 5, 6)$
The system of equations are now,

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$$

$\boldsymbol{A}^{-1}$ exists if and only if $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ has only one solution.

Matrix Inversion: Motivations

Consider the following equations:

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ 0x_1 + 5x_2 + 0x_3 &= 5 \\ 0x_1 + 0x_2 + 3x_3 &= 6 \end{aligned}$$

$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$
$\boldsymbol{x} = (x_1, x_2, x_3)$
$\boldsymbol{b} = (0, 5, 6)$
The system of equations are now,

$$\boldsymbol{Ax} = \boldsymbol{b}$$

$\boldsymbol{A}^{-1}$ exists if and only if $\boldsymbol{Ax} = \boldsymbol{b}$ has only one solution.

Matrix Inversion: Motivations

Consider the following equations:

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ 0x_1 + 5x_2 + 0x_3 &= 5 \\ 0x_1 + 0x_2 + 3x_3 &= 6 \end{aligned}$$

$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

$\boldsymbol{x} = (x_1, x_2, x_3)$

$\boldsymbol{b} = (0, 5, 6)$

The system of equations are now,

$$\boldsymbol{Ax} = \boldsymbol{b}$$

$\boldsymbol{A}^{-1}$ exists if and only if $\boldsymbol{Ax} = \boldsymbol{b}$ has only one solution.

Matrix Inversion: Motivations

Consider the following equations:

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ 0x_1 + 5x_2 + 0x_3 &= 5 \\ 0x_1 + 0x_2 + 3x_3 &= 6 \end{aligned}$$

$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$
$\boldsymbol{x} = (x_1, x_2, x_3)$
$\boldsymbol{b} = (0, 5, 6)$
The system of equations are now,

$$\boldsymbol{Ax} = \boldsymbol{b}$$

$\boldsymbol{A}^{-1}$ exists if and only if$\boldsymbol{Ax} = \boldsymbol{b}$ has only one solution.

Matrix Inversion, Definition

Definition

Suppose $\boldsymbol{X}$ is an $n \times n$ matrix. We will call $\boldsymbol{X}^{-1}$ the inverse of $\boldsymbol{X}$ if

$$\boldsymbol{X}^{-1}\boldsymbol{X} \;=\; \boldsymbol{X}\boldsymbol{X}^{-1} = \boldsymbol{I}$$

If $\boldsymbol{X}^{-1}$ exists then $\boldsymbol{X}$ is invertible. If $\boldsymbol{X}^{-1}$ does not exist, then we will say $\boldsymbol{X}$ is singular.

Matrix Inversion

Now, let's discuss some properties of matrix inversion.

Theorem

The inverse of matrix $\boldsymbol{X}$, $\boldsymbol{X}^{-1}$, is unique

Proof.

By way of contradiction, suppose not. Then there are at least two matrices $\boldsymbol{A}$ and $\boldsymbol{C}$ such that $\boldsymbol{AX} = \boldsymbol{I}$ and $\boldsymbol{CX} = \boldsymbol{I}$

This implies that,

$$\begin{aligned} \boldsymbol{AXC} &= (\boldsymbol{AX})C \\ &= \boldsymbol{IC} \\ &= \boldsymbol{C} \end{aligned}$$

Matrix Inversion

Now, let's discuss some properties of matrix inversion.

Theorem

The inverse of matrix $\boldsymbol{X}$, $\boldsymbol{X}^{-1}$, is unique

Proof.

By way of contradiction, suppose not. Then there are at least two matrices $\boldsymbol{A}$ and $\boldsymbol{C}$ such that $\boldsymbol{AX} = \boldsymbol{I}$ and $\boldsymbol{CX} = \boldsymbol{I}$

This implies that,

$$\begin{aligned} \boldsymbol{AXC} &= (\boldsymbol{AX})C \\ &= \boldsymbol{IC} \\ &= \boldsymbol{C} \end{aligned}$$

Matrix Inversion

Now, let's discuss some properties of matrix inversion.

Theorem

The inverse of matrix $\boldsymbol{X}$, $\boldsymbol{X}^{-1}$, is unique

Proof.

By way of contradiction, suppose not. Then there are at least two matrices $\boldsymbol{A}$ and $\boldsymbol{C}$ such that $\boldsymbol{AX} = \boldsymbol{I}$ and $\boldsymbol{CX} = \boldsymbol{I}$

This implies that,

$$\begin{aligned} \boldsymbol{AXC} &= (\boldsymbol{AX})C \\ &= \boldsymbol{IC} \\ &= \boldsymbol{C} \end{aligned}$$

Matrix Inversion

Now, let's discuss some properties of matrix inversion.

Theorem

The inverse of matrix $\boldsymbol{X}$, $\boldsymbol{X}^{-1}$, is unique

Proof.

By way of contradiction, suppose not. Then there are at least two matrices $\boldsymbol{A}$ and $\boldsymbol{C}$ such that $\boldsymbol{AX} = \boldsymbol{I}$ and $\boldsymbol{CX} = \boldsymbol{I}$

This implies that,

$$\begin{aligned} \boldsymbol{AXC} &= (\boldsymbol{AX})C \\ &= \boldsymbol{IC} \\ &= \boldsymbol{C} \end{aligned}$$

Matrix Inversion

Now, let's discuss some properties of matrix inversion.

Theorem

The inverse of matrix $\boldsymbol{X}$, $\boldsymbol{X}^{-1}$, is unique

Proof.
By way of contradiction, suppose not. Then there are at least two matrices $\boldsymbol{A}$ and $\boldsymbol{C}$ such that $\boldsymbol{AX} = \boldsymbol{I}$ and $\boldsymbol{CX} = \boldsymbol{I}$
This implies that,

$$\begin{aligned} \boldsymbol{AXC} &= (\boldsymbol{AX})C \\ &= \boldsymbol{IC} \\ &= \boldsymbol{C} \end{aligned}$$

Matrix Inversion

Now, let's discuss some properties of matrix inversion.

Theorem

The inverse of matrix $\boldsymbol{X}$, $\boldsymbol{X}^{-1}$, is unique

Proof.

By way of contradiction, suppose not. Then there are at least two matrices $\boldsymbol{A}$ and $\boldsymbol{C}$ such that $\boldsymbol{AX} = \boldsymbol{I}$ and $\boldsymbol{CX} = \boldsymbol{I}$

This implies that,

$$\begin{aligned} \boldsymbol{AXC} &= (\boldsymbol{AX})C \\ &= \boldsymbol{IC} \\ &= \boldsymbol{C} \end{aligned}$$

Matrix Inversion

Now, let's discuss some properties of matrix inversion.

Theorem

The inverse of matrix $\boldsymbol{X}$, $\boldsymbol{X}^{-1}$, is unique

Proof.

By way of contradiction, suppose not. Then there are at least two matrices $\boldsymbol{A}$ and $\boldsymbol{C}$ such that $\boldsymbol{AX} = \boldsymbol{I}$ and $\boldsymbol{CX} = \boldsymbol{I}$

This implies that,

$$\begin{aligned}\boldsymbol{AXC} &= (\boldsymbol{AX})C \\ &= \boldsymbol{IC} \\ &= \boldsymbol{C}\end{aligned}$$

Matrix Inversion

But it also implies that

$$\begin{aligned} \boldsymbol{AXC} &= \boldsymbol{A}(\boldsymbol{XC}) \\ &= \boldsymbol{A}(\boldsymbol{I}) \\ &= \boldsymbol{A} \end{aligned}$$

So $\boldsymbol{C} = \boldsymbol{AXC} = \boldsymbol{A}$ or $\boldsymbol{C} = \boldsymbol{A}$ but this contradicts our assumption that there are two unique inverses.

Matrix Inversion

But it also implies that

$$\begin{aligned} \boldsymbol{AXC} &= \boldsymbol{A}(\boldsymbol{XC}) \\ &= \boldsymbol{A}(\boldsymbol{I}) \\ &= \boldsymbol{A} \end{aligned}$$

So $\boldsymbol{C} = \boldsymbol{AXC} = \boldsymbol{A}$ or $\boldsymbol{C} = \boldsymbol{A}$ but this contradicts our assumption that there are two unique inverses.

Matrix Inversion

But it also implies that

$$
\begin{aligned}
\boldsymbol{AXC} &= \boldsymbol{A}(\boldsymbol{XC}) \\
&= \boldsymbol{A}(\boldsymbol{I}) \\
&= \boldsymbol{A}
\end{aligned}
$$

So $\boldsymbol{C} = \boldsymbol{AXC} = \boldsymbol{A}$ or $\boldsymbol{C} = \boldsymbol{A}$ but this contradicts our assumption that there are two unique inverses.

Matrix Inversion

But it also implies that

$$\begin{aligned} \boldsymbol{AXC} &= \boldsymbol{A}(\boldsymbol{XC}) \\ &= \boldsymbol{A}(\boldsymbol{I}) \\ &= \boldsymbol{A} \end{aligned}$$

So $\boldsymbol{C} = \boldsymbol{AXC} = \boldsymbol{A}$ or $\boldsymbol{C} = \boldsymbol{A}$ but this contradicts our assumption that there are two unique inverses.

Matrix Inversion

But it also implies that

$$\begin{aligned} \boldsymbol{AXC} &= \boldsymbol{A}(\boldsymbol{XC}) \\ &= \boldsymbol{A}(\boldsymbol{I}) \\ &= \boldsymbol{A} \end{aligned}$$

So $\boldsymbol{C} = \boldsymbol{AXC} = \boldsymbol{A}$ or $\boldsymbol{C} = \boldsymbol{A}$ but this contradicts our assumption that there are two unique inverses.

Matrix Inversion

But it also implies that

$$\begin{aligned} \boldsymbol{AXC} &= \boldsymbol{A}(\boldsymbol{XC}) \\ &= \boldsymbol{A}(\boldsymbol{I}) \\ &= \boldsymbol{A} \end{aligned}$$

So $\boldsymbol{C} = \boldsymbol{AXC} = \boldsymbol{A}$ or $\boldsymbol{C} = \boldsymbol{A}$ but this contradicts our assumption that there are two unique inverses.

Matrix Inversion

Theorem

Suppose $\boldsymbol{A}$ *has inverse* $\boldsymbol{A}^{-1}$ *and* $\boldsymbol{B}$ *has inverse* $\boldsymbol{B}^{-1}$. *Then,*

$$(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$$

Proof.

We need to show that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) = (\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) = \boldsymbol{I}$.

$$\begin{aligned}(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) &= \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} \\ &= \boldsymbol{B}^{-1}\boldsymbol{I}\boldsymbol{B} \\ &= \boldsymbol{B}^{-1}\boldsymbol{B} \\ &= \boldsymbol{I}\end{aligned}$$

Matrix Inversion

Theorem

Suppose $\boldsymbol{A}$ *has inverse* $\boldsymbol{A}^{-1}$ *and* $\boldsymbol{B}$ *has inverse* $\boldsymbol{B}^{-1}$*. Then,*

$$(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$$

Proof.

We need to show that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) = (\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) = \boldsymbol{I}$.

$$\begin{aligned}
(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) &= \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} \\
&= \boldsymbol{B}^{-1}\boldsymbol{I}\boldsymbol{B} \\
&= \boldsymbol{B}^{-1}\boldsymbol{B} \\
&= \boldsymbol{I}
\end{aligned}$$

Matrix Inversion

Theorem

Suppose $\boldsymbol{A}$ *has inverse* $\boldsymbol{A}^{-1}$ *and* $\boldsymbol{B}$ *has inverse* $\boldsymbol{B}^{-1}$*. Then,*

$$(\boldsymbol{AB})^{-1} \;=\; \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$$

Proof.

We need to show that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) = (\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) = \boldsymbol{I}$.

$$\begin{aligned}
(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) &= \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} \\
&= \boldsymbol{B}^{-1}\boldsymbol{I}\boldsymbol{B} \\
&= \boldsymbol{B}^{-1}\boldsymbol{B} \\
&= \boldsymbol{I}
\end{aligned}$$

Matrix Inversion

Theorem

Suppose $\boldsymbol{A}$ has inverse $\boldsymbol{A}^{-1}$ and $\boldsymbol{B}$ has inverse $\boldsymbol{B}^{-1}$. Then,

$$(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$$

Proof.
We need to show that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) = (\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) = \boldsymbol{I}$.

$$\begin{aligned}(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) &= \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} \\ &= \boldsymbol{B}^{-1}\boldsymbol{I}\boldsymbol{B} \\ &= \boldsymbol{B}^{-1}\boldsymbol{B} \\ &= \boldsymbol{I}\end{aligned}$$

Matrix Inversion

Theorem

Suppose $\boldsymbol{A}$ has inverse $\boldsymbol{A}^{-1}$ and $\boldsymbol{B}$ has inverse $\boldsymbol{B}^{-1}$. Then,

$$(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$$

Proof.
We need to show that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) = (\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) = \boldsymbol{I}$.

$$\begin{aligned}
(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) &= \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} \\
&= \boldsymbol{B}^{-1}\boldsymbol{I}\boldsymbol{B} \\
&= \boldsymbol{B}^{-1}\boldsymbol{B} \\
&= \boldsymbol{I}
\end{aligned}$$

Matrix Inversion

Theorem

Suppose $\boldsymbol{A}$ has inverse $\boldsymbol{A}^{-1}$ and $\boldsymbol{B}$ has inverse $\boldsymbol{B}^{-1}$. Then,

$$(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$$

Proof.

We need to show that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) = (\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) = \boldsymbol{I}$.

$$\begin{aligned}
(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{AB}) &= \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} \\
&= \boldsymbol{B}^{-1}\boldsymbol{I}\boldsymbol{B} \\
&= \boldsymbol{B}^{-1}\boldsymbol{B} \\
&= \boldsymbol{I}
\end{aligned}$$

Matrix Inversion

$$\begin{aligned}(\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) &= \boldsymbol{A}(\boldsymbol{B}\boldsymbol{B}^{-1})\boldsymbol{A}^{-1}\\ &= \boldsymbol{A}\boldsymbol{I}\boldsymbol{A}^{-1}\\ &= \boldsymbol{A}\boldsymbol{A}^{-1}\\ &= \boldsymbol{I}\end{aligned}$$

So $\boldsymbol{AB}$ is invertible and $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.

Matrix Inversion

$$
\begin{aligned}
(\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) &= \boldsymbol{A}(\boldsymbol{B}\boldsymbol{B}^{-1})\boldsymbol{A}^{-1} \\
&= \boldsymbol{A}\boldsymbol{I}\boldsymbol{A}^{-1} \\
&= \boldsymbol{A}\boldsymbol{A}^{-1} \\
&= \boldsymbol{I}
\end{aligned}
$$

So $\boldsymbol{AB}$ is invertible and $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.

Matrix Inversion

$$\begin{aligned}(\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) &= \boldsymbol{A}(\boldsymbol{BB}^{-1})\boldsymbol{A}^{-1} \\ &= \boldsymbol{AIA}^{-1} \\ &= \boldsymbol{AA}^{-1} \\ &= \boldsymbol{I}\end{aligned}$$

So $\boldsymbol{AB}$ is invertible and $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.

Matrix Inversion

$$\begin{aligned}(\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) &= \boldsymbol{A}(\boldsymbol{BB}^{-1})\boldsymbol{A}^{-1}\\ &= \boldsymbol{AIA}^{-1}\\ &= \boldsymbol{AA}^{-1}\\ &= \boldsymbol{I}\end{aligned}$$

So $\boldsymbol{AB}$ is invertible and $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.

Matrix Inversion

$$
\begin{aligned}
(\boldsymbol{AB})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) &= \boldsymbol{A}(\boldsymbol{BB}^{-1})\boldsymbol{A}^{-1} \\
&= \boldsymbol{AIA}^{-1} \\
&= \boldsymbol{AA}^{-1} \\
&= \boldsymbol{I}
\end{aligned}
$$

So $\boldsymbol{AB}$ is invertible and $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.

Inclass Exercise: Challenge Inversion Proofs

- Show that $(\boldsymbol{A}^{-1})^{-1} = \boldsymbol{A}$.
- Show that $(k\boldsymbol{A})^{-1} = \frac{1}{k}\boldsymbol{A}^{-1}$

Matrix Inversion

After we define matrix inversion and discuss its properties, a natural question to ask is that when do matrix inversion exist?

The answer relates to a concept, linear independence.

Intuitively, linear independence means that there is no repeated information in the matrix.

Matrix Inversion

After we define matrix inversion and discuss its properties, a natural question to ask is that when do matrix inversion exist?

The answer relates to a concept, linear independence.

Intuitively, linear independence means that there is no repeated information in the matrix.

Matrix Inversion

After we define matrix inversion and discuss its properties, a natural question to ask is that when do matrix inversion exist?

The answer relates to a concept, linear independence.

Intuitively, linear independence means that there is no repeated information in the matrix.

Matrix Inversion: Existence

Definition

Suppose we have a set of vectors $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_r\}$
And consider the system of equations

$$k_1\boldsymbol{v}_1 + k_2\boldsymbol{v}_2 + \ldots + k_r\boldsymbol{v}_r = \mathbf{0}$$

If the only solution is $k_1 = 0, k_2 = 0, k_3 = 0, \ldots, k_r = 0$ *then we say that the set is linearly independent. If there are other solutions, then the set is linearly dependent.*

Matrix Inversion: Existence

Consider $\boldsymbol{v}_1 = (1, 0, 0)$, $\boldsymbol{v}_2 = (0, 1, 0)$, $\boldsymbol{v}_3 = (0, 0, 1)$
Can we write this as a combination of other vectors? no!
Consider $\boldsymbol{v}_1 = (1, 0, 0)$, $\boldsymbol{v}_2 = (0, 1, 0)$, $\boldsymbol{v}_3 = (0, 0, 1)$, $\boldsymbol{v}_4 = (1, 2, 3)$.
Can we write any of these vector as a combination of the remaining two vectors?

$$\boldsymbol{v}_4 = \boldsymbol{v}_1 + 2\boldsymbol{v}_2 + 3\boldsymbol{v}_3$$

Matrix Inversion: Existence

Consider $\boldsymbol{v}_1 = (1,0,0)$, $\boldsymbol{v}_2 = (0,1,0)$, $\boldsymbol{v}_3 = (0,0,1)$
Can we write this as a combination of other vectors? no!
Consider $\boldsymbol{v}_1 = (1,0,0)$, $\boldsymbol{v}_2 = (0,1,0)$, $\boldsymbol{v}_3 = (0,0,1)$, $\boldsymbol{v}_4 = (1,2,3)$.
Can we write any of these vector as a combination of the remaining two vectors?

$$\boldsymbol{v}_4 = \boldsymbol{v}_1 + 2\boldsymbol{v}_2 + 3\boldsymbol{v}_3$$

Matrix Inversion: Existence

Consider $\boldsymbol{v}_1 = (1, 0, 0)$, $\boldsymbol{v}_2 = (0, 1, 0)$, $\boldsymbol{v}_3 = (0, 0, 1)$
Can we write this as a combination of other vectors? no!
Consider $\boldsymbol{v}_1 = (1, 0, 0)$, $\boldsymbol{v}_2 = (0, 1, 0)$, $\boldsymbol{v}_3 = (0, 0, 1)$, $\boldsymbol{v}_4 = (1, 2, 3)$.
Can we write any of these vector as a combination of the remaining two vectors?

$$\boldsymbol{v}_4 = \boldsymbol{v}_1 + 2\boldsymbol{v}_2 + 3\boldsymbol{v}_3$$

Matrix Inversion: Existence

Consider $\boldsymbol{v}_1 = (1, 0, 0)$, $\boldsymbol{v}_2 = (0, 1, 0)$, $\boldsymbol{v}_3 = (0, 0, 1)$
Can we write this as a combination of other vectors? no!
Consider $\boldsymbol{v}_1 = (1, 0, 0)$, $\boldsymbol{v}_2 = (0, 1, 0)$, $\boldsymbol{v}_3 = (0, 0, 1)$, $\boldsymbol{v}_4 = (1, 2, 3)$.
Can we write any of these vector as a combination of the remaining two vectors?

$$\boldsymbol{v}_4 = \boldsymbol{v}_1 + 2\boldsymbol{v}_2 + 3\boldsymbol{v}_3$$

Matrix Inversion: Existence

Theorem

Suppose $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_K \in R^n$. If $K > n$ then the set is linearly dependent

- $\boldsymbol{v}_1 = (v_{11}, v_{21}, \ldots, v_{n1})$
- Says that if there are more vectors in the set than elements in each vector, one must be linearly dependent

Matrix Inversion: Existence

Theorem

Suppose $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_K \in R^n$. If $K > n$ then the set is linearly dependent

- $\boldsymbol{v}_1 = (v_{11}, v_{21}, \ldots, v_{n1})$
- Says that if there are more vectors in the set than elements in each vector, one must be linearly dependent

Matrix Inversion: Existence

Theorem

Suppose $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_K \in R^n$. *If* $K > n$ *then the set is linearly dependent*

- $\boldsymbol{v}_1 = (v_{11}, v_{21}, \ldots, v_{n1})$
- Says that if there are more vectors in the set than elements in each vector, one must be linearly dependent

Matrix Inversion: Existence

The following theorem tells that

Theorem

Suppose $\boldsymbol{X}$ is an $n \times n$ matrix. Recall we can write this matrix as $\begin{pmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \\ \vdots \\ \boldsymbol{x}_n \end{pmatrix}$.

Then $\boldsymbol{X}$ has an inverse if and only if $S = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\}$ is linearly independent

If this is true, we say $\boldsymbol{X}$ has full rank.

Matrix Inversion: Existence

The following theorem tells that

Theorem

Suppose $\boldsymbol{X}$ *is an* $n \times n$ *matrix. Recall we can write this matrix as* $\begin{pmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \\ \vdots \\ \boldsymbol{x}_n \end{pmatrix}$.

Then $\boldsymbol{X}$ *has an inverse if and only if* $S = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\}$ *is linearly independent*

If this is true, we say $\boldsymbol{X}$ has full rank.

Matrix Inversion: Rank and Span

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Columns 1 and 2 are linearly independent, but their sum equals Column 3; the rank of $\mathbf{A} = 2$

These three vectors span $\mathbb{R}^2$ but not $\mathbb{R}^3$: Show that for any $(x_1, x_2) \in \mathbb{R}^2$, we can find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1,0,1) + \lambda_2(0,1,1) + \lambda_3(1,1,2) = (x_1, x_2, x_1 + x_2)$$

They span $\mathbb{R}^2$ and the space of all vectors in $\mathbb{R}^3$ of form $(x_1, x_2, x_1 + x_2)$

Show that for any $(x_1, x_2, x_3) \in \mathbb{R}^3$, we *can't* find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1,0,1) + \lambda_2(0,1,1) + \lambda_3(1,1,2) = (x_1, x_2, x_3)$$

Matrix Inversion: Rank and Span

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Columns 1 and 2 are linearly independent, but their sum equals Column 3; the rank of $\mathbf{A} = 2$

These three vectors span $\mathbb{R}^2$ but not $\mathbb{R}^3$: Show that for any $(x_1, x_2) \in \mathbb{R}^2$, we can find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1,0,1) + \lambda_2(0,1,1) + \lambda_3(1,1,2) = (x_1, x_2, x_1 + x_2)$$

They span $\mathbb{R}^2$ and the space of all vectors in $\mathbb{R}^3$ of form $(x_1, x_2, x_1 + x_2)$

Show that for any $(x_1, x_2, x_3) \in \mathbb{R}^3$, we *can't* find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1,0,1) + \lambda_2(0,1,1) + \lambda_3(1,1,2) = (x_1, x_2, x_3)$$

Matrix Inversion: Rank and Span

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Columns 1 and 2 are linearly independent, but their sum equals Column 3; the rank of $\mathbf{A} = 2$

These three vectors span $\mathbb{R}^2$ but not $\mathbb{R}^3$: Show that for any $(x_1, x_2) \in \mathbb{R}^2$, we can find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1,0,1) + \lambda_2(0,1,1) + \lambda_3(1,1,2) = (x_1, x_2, x_1 + x_2)$$

They span $\mathbb{R}^2$ and the space of all vectors in $\mathbb{R}^3$ of form $(x_1, x_2, x_1 + x_2)$

Show that for any $(x_1, x_2, x_3) \in \mathbb{R}^3$, we *can't* find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1,0,1) + \lambda_2(0,1,1) + \lambda_3(1,1,2) = (x_1, x_2, x_3)$$

Matrix Inversion: Rank and Span

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Columns 1 and 2 are linearly independent, but their sum equals Column 3; the rank of $\mathbf{A} = 2$

These three vectors span $\mathbb{R}^2$ but not $\mathbb{R}^3$: Show that for any $(x_1, x_2) \in \mathbb{R}^2$, we can find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1,0,1) + \lambda_2(0,1,1) + \lambda_3(1,1,2) = (x_1, x_2, x_1 + x_2)$$

They span $\mathbb{R}^2$ and the space of all vectors in $\mathbb{R}^3$ of form $(x_1, x_2, x_1 + x_2)$

Show that for any $(x_1, x_2, x_3) \in \mathbb{R}^3$, we *can't* find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1,0,1) + \lambda_2(0,1,1) + \lambda_3(1,1,2) = (x_1, x_2, x_3)$$

Matrix Inversion: Rank and Span

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Columns 1 and 2 are linearly independent, but their sum equals Column 3; the rank of $\mathbf{A} = 2$

These three vectors span $\mathbb{R}^2$ but not $\mathbb{R}^3$: Show that for any $(x_1, x_2) \in \mathbb{R}^2$, we can find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1, 0, 1) + \lambda_2(0, 1, 1) + \lambda_3(1, 1, 2) = (x_1, x_2, x_1 + x_2)$$

They span $\mathbb{R}^2$ and the space of all vectors in $\mathbb{R}^3$ of form $(x_1, x_2, x_1 + x_2)$

Show that for any $(x_1, x_2, x_3) \in \mathbb{R}^3$, we *can't* find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1, 0, 1) + \lambda_2(0, 1, 1) + \lambda_3(1, 1, 2) = (x_1, x_2, x_3)$$

Matrix Inversion: Rank and Span

Finding the rank of a matrix **A**

- Reduce **A** to an echelon matrix **E** using Gaussian elimination
- The number of non-zero rows of **E** is the rank of **A**!

Properties of ranks

- The rank of an $n \times m$ matrix **A** can't be greater than either n or m: Rank(**A**)$\leq \min(n, m)$
- A matrix has *full rank* if its rank is the biggest it could be: Rank(**A**)$= \min(n, m)$
- A square matrix is *invertible* if and only if it has full rank

Matrix Inversion: Rank and Span

Finding the rank of a matrix $\mathbf{A}$

- Reduce $\mathbf{A}$ to an echelon matrix $\mathbf{E}$ using Gaussian elimination
- The number of non-zero rows of $\mathbf{E}$ is the rank of $\mathbf{A}$!

Properties of ranks

- The rank of an $n \times m$ matrix $\mathbf{A}$ can't be greater than either n or m: $\text{Rank}(\mathbf{A}) \leq \min(n, m)$
- A matrix has *full rank* if its rank is the biggest it could be: $\text{Rank}(\mathbf{A}) = \min(n, m)$
- A square matrix is *invertible* if and only if it has full rank

Matrix Inversion: Rank and Span

Finding the rank of a matrix $\mathbf{A}$

- Reduce $\mathbf{A}$ to an echelon matrix $\mathbf{E}$ using Gaussian elimination
- The number of non-zero rows of $\mathbf{E}$ is the rank of $\mathbf{A}$!

Properties of ranks

- The rank of an $n \times m$ matrix $\mathbf{A}$ can't be greater than either n or m: $\text{Rank}(\mathbf{A}) \leq \min(n, m)$
- A matrix has *full rank* if its rank is the biggest it could be: $\text{Rank}(\mathbf{A}) = \min(n, m)$
- A square matrix is *invertible* if and only if it has full rank

Matrix Inversion: Rank and Span

Finding the rank of a matrix $\mathbf{A}$

- Reduce $\mathbf{A}$ to an echelon matrix $\mathbf{E}$ using Gaussian elimination
- The number of non-zero rows of $\mathbf{E}$ is the rank of $\mathbf{A}$!

Properties of ranks

- The rank of an $n \times m$ matrix $\mathbf{A}$ can't be greater than either n or m: Rank($\mathbf{A}$)$\leq \min(n, m)$
- A matrix has *full rank* if its rank is the biggest it could be: Rank($\mathbf{A}$)$= \min(n, m)$
- A square matrix is *invertible* if and only if it has full rank

Matrix Inversion: Rank and Span

Finding the rank of a matrix **A**

- Reduce **A** to an echelon matrix **E** using Gaussian elimination
- The number of non-zero rows of **E** is the rank of **A**!

Properties of ranks

- The rank of an $n \times m$ matrix **A** can't be greater than either n or m: Rank($\mathbf{A}$)$\leq \min(n, m)$
- A matrix has *full rank* if its rank is the biggest it could be: Rank($\mathbf{A}$)$= \min(n, m)$
- A square matrix is *invertible* if and only if it has full rank

Matrix Inversion: Rank and Span

Finding the rank of a matrix $\mathbf{A}$

- Reduce $\mathbf{A}$ to an echelon matrix $\mathbf{E}$ using Gaussian elimination
- The number of non-zero rows of $\mathbf{E}$ is the rank of $\mathbf{A}$!

Properties of ranks

- The rank of an $n \times m$ matrix $\mathbf{A}$ can't be greater than either n or m: Rank($\mathbf{A}$)$\leq \min(n, m)$
- A matrix has *full rank* if its rank is the biggest it could be: Rank($\mathbf{A}$)$= \min(n, m)$
- A square matrix is *invertible* if and only if it has full rank

Matrix Inversion: Rank and Span

Finding the rank of a matrix $\mathbf{A}$

- Reduce $\mathbf{A}$ to an echelon matrix $\mathbf{E}$ using Gaussian elimination
- The number of non-zero rows of $\mathbf{E}$ is the rank of $\mathbf{A}$!

Properties of ranks

- The rank of an $n \times m$ matrix $\mathbf{A}$ can't be greater than either n or m: Rank($\mathbf{A}$)$\leq \min(n, m)$
- A matrix has *full rank* if its rank is the biggest it could be: Rank($\mathbf{A}$)$= \min(n, m)$
- A square matrix is *invertible* if and only if it has full rank

Matrix Inversion: Gauss-Jordan elimination

How to invert a matrix

Take

$$[\mathbf{AI}]$$

and use Gauss-Jordan elimination to convert it into

$$[\mathbf{IA}^{-1}]$$

Gauss-Jordan elimination is similar to Gaussian elimination, but turns matrix into a diagonal matrix, instead of just an echelon matrix

Matrix Inversion: Gauss-Jordan elimination

How to invert a matrix

Take

$$\begin{bmatrix}\mathbf{A}\mathbf{I}\end{bmatrix}$$

and use Gauss-Jordan elimination to convert it into

$$\begin{bmatrix}\mathbf{I}\mathbf{A}^{-1}\end{bmatrix}$$

Gauss-Jordan elimination is similar to Gaussian elimination, but turns matrix into a diagonal matrix, instead of just an echelon matrix

Matrix Inversion: Gauss-Jordan elimination

How to invert a matrix

Take

$$\begin{bmatrix}\mathbf{AI}\end{bmatrix}$$

and use Gauss-Jordan elimination to convert it into

$$\begin{bmatrix}\mathbf{IA^{-1}}\end{bmatrix}$$

Gauss-Jordan elimination is similar to Gaussian elimination, but turns matrix into a diagonal matrix, instead of just an echelon matrix

Matrix Inversion: Gauss-Jordan elimination

Let's invert

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 1 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

$$\text{Step 1: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Step 2: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Matrix Inversion: Gauss-Jordan elimination

Let's invert

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 1 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

$$\text{Step 1: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Step 2: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Matrix Inversion: Gauss-Jordan elimination

Let's invert

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 1 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

$$\text{Step 1:} \quad \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Step 2:} \quad \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 2:} \quad \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Since we are creating a diagonal we want to get rid of the "1" in (1, 2)

$$\text{Step 3:} \quad \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

This added $2 \times$ Row 2 to Row 1 and subtracted Row 2 from Row 3

Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 2:} \quad \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Since we are creating a diagonal we want to get rid of the “1” in (1, 2)

$$\text{Step 3:} \quad \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

This added 2 $\times$ Row 2 to Row 1 and subtracted Row 2 from Row 3

Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 2: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Since we are creating a diagonal we want to get rid of the "1" in (1, 2)

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

This added 2 $\times$ Row 2 to Row 1 and subtracted Row 2 from Row 3

Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We get a diagonal matrix by adding $2 \times$ Row 3 to Rows 1 & 2

$$\text{Step 4: } \begin{bmatrix} 2 & 0 & 0 & -2 & 0 & 2 \\ 0 & -\frac{1}{2} & 0 & -\frac{3}{2} & -1 & 2 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We finish by normalizing

$$\text{Step 5: } \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 2 & -4 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 2 & -4 \\ 0 & -1 & 1 \end{bmatrix}$$

Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We get a diagonal matrix by adding $2 \times$ Row 3 to Rows 1 & 2

$$\text{Step 4: } \begin{bmatrix} 2 & 0 & 0 & -2 & 0 & 2 \\ 0 & -\frac{1}{2} & 0 & -\frac{3}{2} & -1 & 2 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We finish by normalizing

$$\text{Step 5: } \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 2 & -4 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 2 & -4 \\ 0 & -1 & 1 \end{bmatrix}$$

Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 3:} \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We get a diagonal matrix by adding $2 \times$ Row 3 to Rows 1 & 2

$$\text{Step 4:} \begin{bmatrix} 2 & 0 & 0 & -2 & 0 & 2 \\ 0 & -\frac{1}{2} & 0 & -\frac{3}{2} & -1 & 2 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We finish by normalizing

$$\text{Step 5:} \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 2 & -4 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 2 & -4 \\ 0 & -1 & 1 \end{bmatrix}$$

Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We get a diagonal matrix by adding $2 \times$ Row 3 to Rows 1 & 2

$$\text{Step 4: } \begin{bmatrix} 2 & 0 & 0 & -2 & 0 & 2 \\ 0 & -\frac{1}{2} & 0 & -\frac{3}{2} & -1 & 2 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We finish by normalizing

$$\text{Step 5: } \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 2 & -4 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 2 & -4 \\ 0 & -1 & 1 \end{bmatrix}$$

Matrix Inversion: An Application

Suppose we have following system of equations:

$$2x + 1y + 2z = 1$$

$$3x + 1y + 1z = 3$$

$$3x + 1y + 2z = 5$$

Say, $\boldsymbol{x} = (x, y, z)^{'}$, $\boldsymbol{b} = (1, 3, 5)^{'}$.

Say, the coefficient matrix for unknowns is A, the inverse of A is computed by the previous example.

Then, we can solve the unknowns by:

$$\boldsymbol{x} = A^{-1}\boldsymbol{b} = (4, -11, 2)^{'}$$